

EmoLab: Synchronized Multimodal Affective Analysis and Replay

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github.com/pawelborowiec-research/EmoLab-Demo

Abstract—EmoLab is an interactive research environment for synchronized inspection, replay, and interpretation of multimodal affect-related model outputs. It organizes audiovisual material, speaker-diarized transcripts, facial observations, prosodic predictions, linguistic predictions, vocal events, timelines, graphs, and structured interpretations within a shared temporal representation. Rather than treating model outputs as direct measurements of a person’s internal emotional state, EmoLab makes those outputs inspectable, replayable, comparable, and challengeable in their temporal and multimodal context. This extended abstract describes the system, its demonstration workflow, and its explicit treatment of provenance, uncertainty, cross-modal disagreement, and interpretive limitations.

Index Terms—affective computing, multimodal interaction, synchronized replay, model inspection, temporal analysis

I. INTRODUCTION AND MOTIVATION

Affective computing encompasses computational approaches for representing, recognizing, and responding to affect-related signals [1]. Multimodal affective-computing systems commonly produce heterogeneous outputs for speech, language, prosody, vocal events, speakers, faces, and temporal interaction structure [2]. These outputs can differ in segmentation, time scale, identifiers, data format, and model semantics, reflecting broader multimodal alignment and representation challenges [2]. Consequently, researchers may find it difficult to reconstruct which model output corresponded to a particular word, speaker turn, facial observation, or moment in the source interaction.

EmoLab addresses this inspection problem by organizing preserved multimodal telemetry and source media into persistent, replayable sessions. The system does not claim to reveal a person’s true internal emotional state, since observable behaviour does not provide an unambiguous readout of internal emotion [3]. Its purpose is to support critical inspection of model-generated affect-related evidence.

II. SYSTEM OVERVIEW

EmoLab maps heterogeneous session elements onto a shared temporal representation. The interface links audiovisual playback with transcript highlighting, speaker information, face-oriented observations, modality-specific affect views, timeline navigation, trajectory maps, EmoTextGraph, and session-level reporting.

The current prototype supports deterministic replay from a locally stored session bundle. Previously generated model outputs can therefore be inspected without repeating the original

inference process or depending on the continued availability of an external inference service.

Figure 1 presents selected interface views used during synchronized multimodal inspection.

III. TECHNICAL CONTRIBUTION

The system provides an integrated inspection workflow centered on:

- temporal alignment of heterogeneous multimodal outputs;
- deterministic replay of preserved source media and telemetry;
- navigation between analytical views and source moments;
- comparison of language, prosody, vocal-event, speaker, and facial information;
- explicit separation of observed model output, cross-channel evidence, interpretive hypotheses, confidence, and uncertainty.

The contribution is not a new emotion-classification model. It is an auditable environment for examining how multiple models behave together across time.

IV. DEMONSTRATION SCENARIO

The demonstration uses a short, previously analysed dyadic audiovisual interaction. The presenter opens a local session, replays the source interaction, inspects the speaker-diarized transcript and facial timeline, navigates to an affect-relevant moment, and compares modality-specific model outputs.

Emotion trajectory maps and EmoTextGraph expose temporal and lexical relations, while the affective-cognitive component organizes selected evidence into confidence-qualified hypotheses, conflicts, and unresolved unknowns.

The project page and demonstration video are available at:

<https://github.com/pawelborowiec-research/EmoLab-Demo>

V. DEMONSTRATION PROTOCOL

The demonstration protocol examines whether a selected source moment can be reconstructed and inspected across linked views. A participant selects a moment, identifies the active speaker and corresponding transcript fragment, compares available modality-specific outputs, and checks whether a higher-level interpretation is temporally grounded in the displayed evidence.

The current paper does not report a formal user study or claim quantitative usability results.

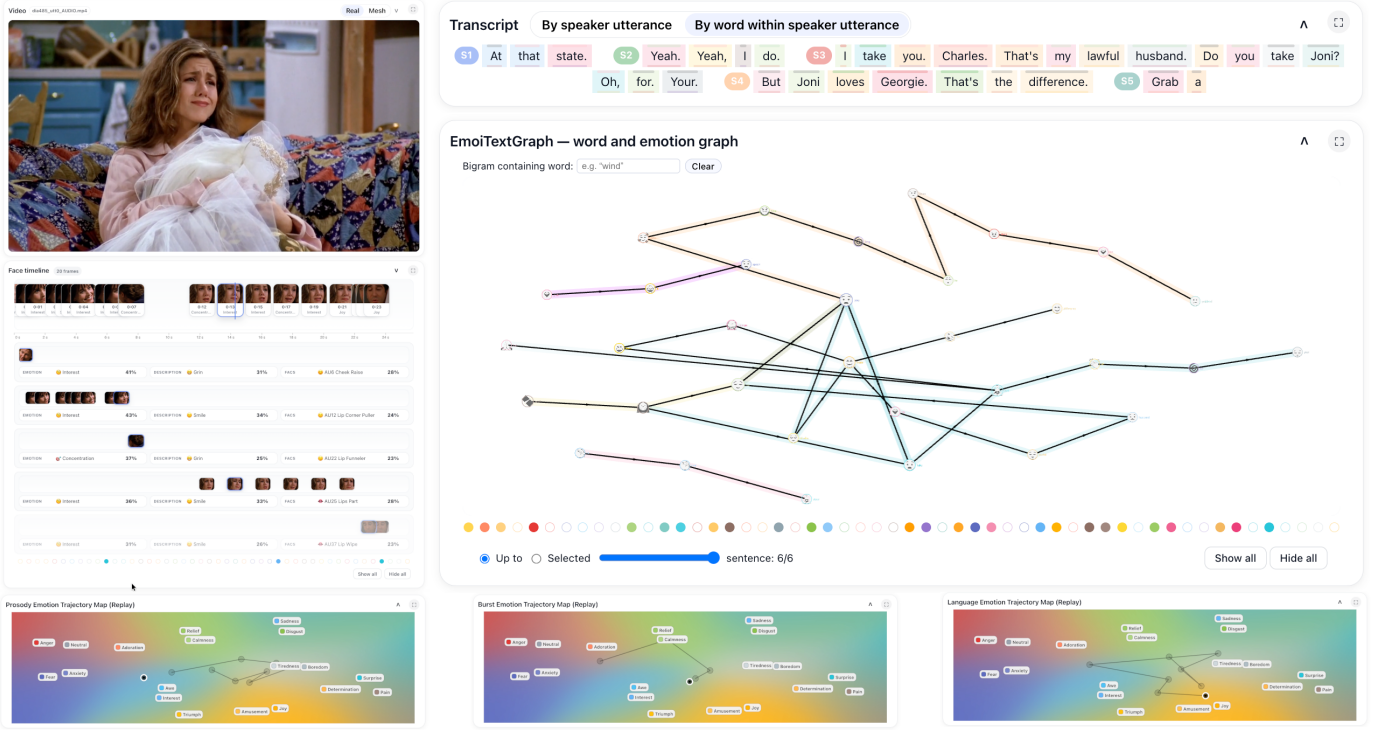


Fig. 1. Selected EmoLab views illustrating synchronized audiovisual replay and face-oriented inspection, speaker-diarized transcript analysis, modality-specific emotion trajectory maps, and EmoTextGraph-based exploration. The views are linked through the shared temporal session representation.

VI. ETHICAL IMPACT AND LIMITATIONS

EmoLab displays probabilistic outputs produced by computational models. Such outputs can be uncertain, culturally dependent, affected by recording conditions, and shaped by dataset or model bias. They must not be interpreted as direct measurements of internal emotional states [3].

The system is not diagnostic and should not be used for consequential decisions concerning health, employment, credibility, psychological condition, access to services, or legal and disciplinary outcomes. Its design instead emphasizes provenance, uncertainty, temporal grounding, and critical human review, consistent with recommendations for responsible automatic emotion-recognition research [4].

VII. CONCLUSION

EmoLab provides a persistent and synchronized environment for inspecting multimodal affect-related telemetry in context. By connecting source media, timelines, transcripts, speakers, faces, modality-specific predictions, and structured interpretation, it supports replayable and challengeable analysis rather than unqualified emotion attribution.

REFERENCES

- [1] R. W. Picard, *Affective Computing*. Cambridge, MA, USA: MIT Press, 1997.
- [2] T. Baltrušaitis, C. Ahuja, and L.-P. Morency, “Multimodal machine learning: A survey and taxonomy,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 41, no. 2, pp. 423–443, 2019.
- [3] L. F. Barrett, R. Adolphs, S. Marsella, A. M. Martinez, and S. D. Pollak, “Emotional expressions reconsidered: Challenges to inferring emotion from human facial movements,” *Psychological Science in the Public Interest*, vol. 20, no. 1, pp. 1–68, 2019.
- [4] S. M. Mohammad, “Ethics sheet for automatic emotion recognition and sentiment analysis,” *Computational Linguistics*, vol. 48, no. 2, pp. 239–278, 2022.